



Time Series Analysis

Chapter 7: Cointegration and VECM



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Acronyms used in this material

- ▣ **ACF**: Autocorrelation Function
- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **AR**: AutoRegressive (model)
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ **DHSY**: Davidson, Hendry, Srba and Yeo (1978)
- ▣ **DOLS**: Dynamic Ordinary Least Squares
- ▣ **DW**: Durbin–Watson (statistic)
- ▣ **ECM**: Error Correction Model
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GDP**: Gross Domestic Product
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PPP**: Purchasing Power Parity
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **ROBOR**: Romanian Interbank Offer Rate
- ▣ **SE**: Standard Error
- ▣ **UK**: United Kingdom
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VECM**: Vector Error Correction Model

Today's question and route

- ▣ **Question:** two prices both wander like random walks; can a combination of them be stable, and how do we model it?
 - ▶ if yes, the series share a common stochastic trend and a long-run equilibrium: regressions in levels are meaningful, not spurious
- ▣ **Route** of the chapter
 - ▶ the idea of cointegration; the Engle–Granger two-step method with the right critical values; error correction models
 - ▶ the VECM and the Granger representation theorem; the Johansen tests; estimating and reading β and α ; forecasting
 - ▶ applications: purchasing power parity, interest rates, three Central European currencies, pairs trading on the Bucharest Stock Exchange
- ▣ We build on [Chapter 3](#) (unit roots, spurious regression) and [Chapter 6](#) (VAR models)

Learning outcomes

- ▣ Define cointegration and explain it as a common stochastic trend
- ▣ Run the Engle–Granger and Phillips–Ouliaris tests and compare them with the right (not the Dickey–Fuller) critical values
- ▣ Estimate an error correction model and interpret the speed of adjustment and the half-life
- ▣ Write a VAR as a VECM, state the Granger representation theorem and choose the cointegration rank with the Johansen tests
- ▣ Interpret cointegrating vectors and adjustment coefficients, and decide when a VECM forecasts better than a VAR in differences
- ▣ Judge an application (parity conditions, pairs trading) out of sample and net of costs

Reading and tools

- ▣ Textbook: [Huang and Petukhina \(2022\)](#), Ch. 9 (nonstationarity and cointegration)
 - ▶ theory: [Hamilton \(1994\)](#), Ch. 19–20; [Lütkepohl \(2005\)](#), Ch. 6–8; the likelihood approach in [Johansen \(1995\)](#) and [Juselius \(2006\)](#)
- ▣ Landmark papers: [Engle and Granger \(1987\)](#), [Johansen \(1988\)](#), [Johansen \(1991\)](#); Granger's Nobel lecture [Granger \(2004\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_07](#)
 - ▶ `coint`, `coint_johansen`, `VECM` and `VAR` from `statsmodels`; `phillips_ouliaris` from `arch`
- ▣ Lecture notebook: [open in Google Colab](#)
- ▣ Video course:  [Applied Time Series Analysis with Python](#)

From spurious regression to cointegration

- Chapter 3: regressing one $I(1)$ series on another, **unrelated** $I(1)$ series gives large t , high R^2 and a Durbin–Watson statistic near 0 Granger and Newbold (1974)
 - ▶ the error $y_t - a - bx_t$ is itself $I(1)$ for every b : nothing ties the two series together
- The exception: for one value of b , the combination $u_t = y_t - a - bx_t$ is **stationary**
 - ▶ then y_t and x_t are **cointegrated** Granger (1981), Engle and Granger (1987): the regression in levels estimates a real long-run relation
 - ▶ the two series may drift apart for a while, but u_t always pulls them back: an **equilibrium**
- Today: how to recognise this case, how to test it, and how to model it

A drunk and her dog

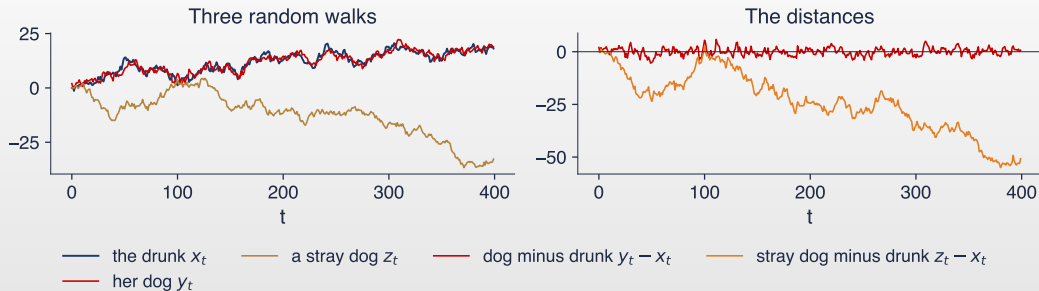


A walk with a dog: two paths that never drift far apart

Photo: Amin (2018); CC BY-SA 4.0; Wikimedia Commons

- Murray (1994): a drunk leaves the pub and wanders: her path x_t is a random walk
 - ▶ her dog also wanders: y_t is a random walk too
 - ▶ the distance between two **independent** walkers grows without limit, like $\sqrt{t}$
- But the dog hears her call and runs back; she hears the dog and turns towards it
 - ▶ each corrects part of the gap: $\Delta x_t = c(y_{t-1} - x_{t-1}) + u_t$, $\Delta y_t = d(x_{t-1} - y_{t-1}) + w_t$; $c, d > 0$: the shares of the gap closed per step; u_t, w_t : independent shocks
 - ▶ both paths remain unpredictable, but the distance $y_t - x_t$ stays bounded: **cointegration** with **error correction**

The drunk and her dog by simulation



- $T = 400$, Gaussian shocks; $c = 0.15$, $d = 0.25$; a stray dog z_t is an independent random walk with the same shocks' variance

Interpreting the simulation: the distances

- Each path is $I(1)$: ADF on x_t gives $p = 0.21$
 - ▶ the distance $y_t - x_t$ is an $AR(1)$ with $\rho = 1 - c - d = 0.6$: ADF $p < 0.001$, standard deviation 1.9
 - ▶ the distance to the stray dog wanders: ADF $p = 0.81$, standard deviation 14.1
- Half of a gap disappears in $\ln 0.5 / \ln 0.6 = 1.4$ steps: the **half-life** (Section 3)
 - ▶ knowing where the drunk is tells us where to look for the dog, even though neither path can be forecast

Definition of cointegration

- **Definition Engle and Granger (1987)**: the components of $\mathbf{y}_t = (y_{1t}, \dots, y_{nt})^\top$ are **cointegrated of order (1,1)**, $\mathbf{y}_t \sim CI(1, 1)$, if
 - ▶ every component is $I(1)$ ([Chapter 3](#): stationary after one difference), and
 - ▶ some linear combination $\beta^\top \mathbf{y}_t$, with $\beta \neq 0$, is $I(0)$: stationary
- β is the **cointegrating vector**; $u_t = \beta^\top \mathbf{y}_t - \mu$ is the **equilibrium error** (μ : its mean; $\top$: transpose)
 - ▶ β is unique only up to scale: 2β works as well; we **normalise** one coefficient to 1, e.g. $\beta = (1, -b)^\top$
 - ▶ with n variables there can be up to $r = n - 1$ independent cointegrating vectors; r is the **cointegration rank**
- Cointegration is a property of the **levels**: differencing the data destroys the information in $\beta^\top \mathbf{y}_t$

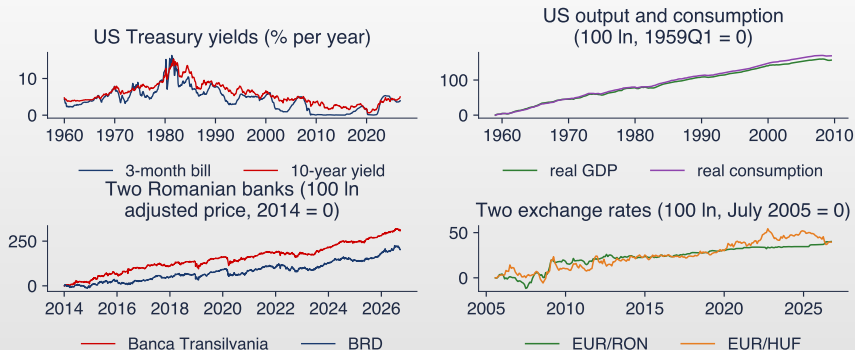
Common stochastic trends

- Write $x_t = x_{t-1} + \varepsilon_t$ (a random walk) and $y_t = b x_t + u_t$, with u_t stationary
 - ▶ both series contain the same stochastic trend $\sum_{s \leq t} \varepsilon_s$; $y_t - b x_t = u_t$ removes it
- **Common trends** representation [Stock and Watson \(1988\)](#): n variables with r cointegrating vectors are driven by $n - r$ common stochastic trends
 - ▶ $r = 0$: n separate trends, no cointegration; $r = n - 1$: one trend shared by all
 - ▶ example: three yields (1, 5 and 10 years) with $r = 2$ share one trend, the **level** of interest rates (Section 5)
- **Worked example**: x_t a random walk, $y_t = 2x_t + u_t$, $z_t = -x_t + v_t$ (u_t, v_t stationary)
 - ▶ $n = 3$, one common trend, so $r = 2$: e.g. $\beta_1 = (1, -2, 0)^\top$ and $\beta_2 = (0, 1, 2)^\top$ (since $y_t + 2z_t = u_t + 2v_t$)

Where economics predicts cointegration

- ▣ **Purchasing power parity:** $s_t = p_t - p_t^* + q_t$, with the real exchange rate q_t stationary (Section 8)
 - ▶ s_t : log exchange rate (lei per euro); p_t, p_t^* : log price levels at home and abroad
- ▣ **Term structure of interest rates:** by the expectations hypothesis, the spread between a long and a short yield is stationary [Campbell and Shiller \(1987\)](#)
- ▣ **Consumption and income:** the “great ratio” $c_t - y_t$ (logs) is stable if consumption follows permanent income
- ▣ **Spot and futures prices,** an index and a fund that tracks it: arbitrage keeps the gap small
- ▣ **Shares of similar firms:** the same sector shocks; the basis of pairs trading (Section 9)

Four pairs of trending series



- US Treasury yields (FRED, monthly, since 1960); US real GDP and consumption (statsmodels macrodata, 1959–2009)
- Banca Transilvania and BRD (adjusted daily prices); EUR/RON and EUR/HUF (BNR reference rates, month-end)

Interpreting the four pairs

- Each series is $I(1)$ (Chapter 3); the question is whether the two series of a pair move **together** in the long run
 - ▶ Engle–Granger test (Section 2), 5% critical value about -3.34
- Yields: $\tau = -3.35$, $p = 0.048$: borderline; consumption and income: $\tau = -3.54$, $p = 0.029$
- The two banks: $\tau = -4.33$, $p = 0.002$: cointegrated in this sample
- EUR/RON and EUR/HUF: $\tau = -2.47$, $p = 0.294$: **not** cointegrated
 - ▶ although they rose by similar amounts and $R^2 = 0.79$ in levels: a common direction is not a common trend

Question for the room

- **Question:** two $I(1)$ series have a correlation of 0.95 in levels. Are they cointegrated?
 - ▶ **Answer:** we cannot tell: unrelated random walks often have high correlations ([Chapter 3](#)); only a test on the equilibrium error decides
- **Question:** two cointegrated series have a correlation of 0.2 between their daily changes. Is this a contradiction?
 - ▶ **Answer:** no: cointegration is about the long run; in the short run the changes can be almost unrelated

Recap: The idea of cointegration

- ▣ $I(1)$ series are cointegrated if a combination $\beta^\top \mathbf{y}_t$ is stationary: a long-run equilibrium
- ▣ Equivalently, they share common stochastic trends: n variables, r vectors, $n - r$ trends
- ▣ Theory suggests cointegration (parity conditions, spreads, great ratios), but the data must confirm it

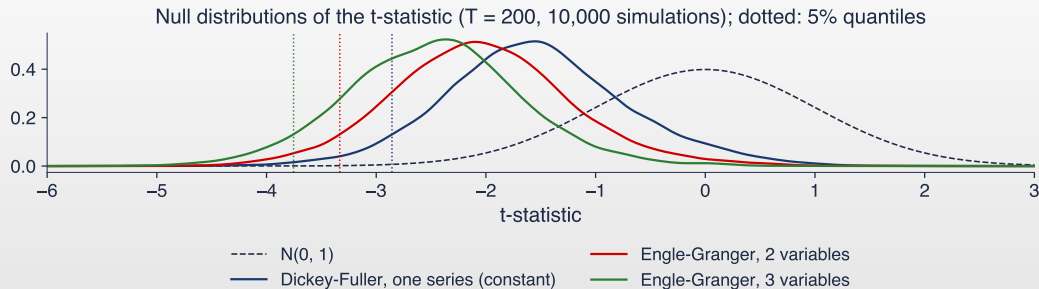
Two steps

- ▣ **Step 1:** estimate the long-run relation by OLS (ordinary least squares) in levels: $y_t = a + b x_t + u_t$
 - ▶ keep the residuals $\hat{u}_t = y_t - \hat{a} - \hat{b}x_t$
 - ▶ if the series are cointegrated, $\hat{b}$ is **super-consistent**: its error shrinks like $1/T$, faster than the usual $1/\sqrt{T}$
- ▣ **Step 2:** an ADF test (Chapter 3) on the residuals, without a constant: $\Delta \hat{u}_t = \gamma \hat{u}_{t-1} + \sum_{j=1}^k \delta_j \Delta \hat{u}_{t-j} + e_t$
 - ▶ $H_0: \gamma = 0$, the residuals have a unit root: **no cointegration**; $H_1: \gamma < 0$: cointegration
 - ▶ δ_j : coefficients of the k lagged differences, which absorb the autocorrelation of $\hat{u}_t$; e_t : white noise
 - ▶ statistic $\tau = \hat{\gamma}/\text{SE}(\hat{\gamma})$, SE: standard error; reject H_0 when τ is below the critical value
- ▣ Pretest: each series must be $I(1)$; cointegration between an $I(0)$ and an $I(1)$ series is impossible

The reason the Dickey–Fuller table is wrong here

- OLS chooses $\hat{b}$ to make the residual variance as small as possible
 - ▶ so $\hat{u}_t$ looks more stationary than the true u_t , even when there is no cointegration
 - ▶ under H_0 the statistic τ is shifted to the left of the Dickey–Fuller distribution
- The critical values depend on the number of variables n and on the deterministic terms of step 1
 - ▶ tables: Engle and Granger (1987) (by simulation), then MacKinnon (1994) and MacKinnon (2010), used by `statsmodels (coint)`
- With β **known** from theory (e.g. a spread, $\beta = (1, -1)^\top$), nothing is estimated: the plain ADF test and its table apply

The null distribution by simulation



- 10 000 samples of $T = 200$ independent Gaussian random walks; step 1 with a constant, step 2 without lags

Interpreting the simulation: the critical values

- 5% quantiles: -2.86 (one series, Dickey–Fuller), -3.33 (2 variables), -3.75 (3 variables)
 - ▶ MacKinnon's values for $T = 200$: -2.88 ; -3.37 ; -3.78 : the simulation reproduces them
- Using the Dickey–Fuller value -2.88 for the residuals would reject a true H_0 in 14.5% of samples with 2 variables and 30.3% with 3
 - ▶ instead of 5%: we would “find” cointegration far too often
- More variables give OLS more freedom to fit, hence more negative critical values

Critical values of the Engle–Granger test

Variables n	constant			constant and trend		
	1%	5%	10%	1%	5%	10%
1 (Dickey–Fuller)	−3.43	−2.86	−2.57	−3.96	−3.41	−3.13
2	−3.90	−3.34	−3.04	−4.33	−3.78	−3.50
3	−4.29	−3.74	−3.45	−4.66	−4.12	−3.84
4	−4.64	−4.10	−3.81	−4.97	−4.43	−4.15

- Asymptotic values ($T \rightarrow \infty$) of [MacKinnon \(2010\)](#); for finite T the values are slightly more negative
- Read the row of the **total** number of variables in the regression, y_t included
- Constant and trend in step 1: when the series have drifts that the relation does not cancel

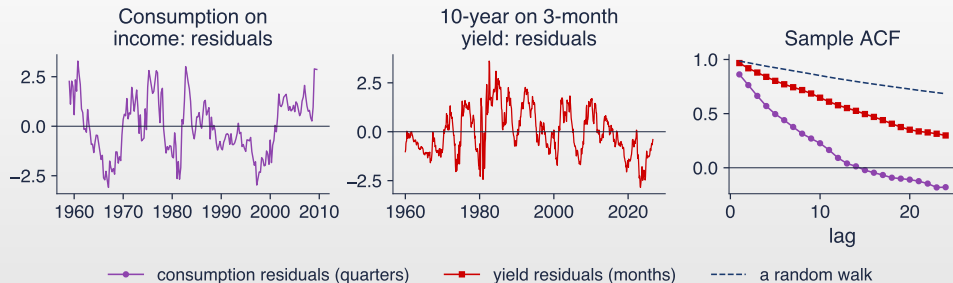
The Phillips–Ouliaris test

- Phillips and Ouliaris (1990): the same step 1 and the same H_0 (no cointegration)
 - ▶ step 2: a Dickey–Fuller regression **without** lagged differences; the t -statistic is corrected with the long-run variance of the residuals (as in the Phillips–Perron test of Chapter 3)
 - ▶ two versions: Z_t (corrected t -ratio) and Z_α (corrected $T\hat{\gamma}$); the critical values of Z_t are those of Engle–Granger
- Reasons for two tests
 - ▶ Engle–Granger depends on the number of lags k chosen by AIC; Phillips–Ouliaris depends on the bandwidth of the long-run variance
 - ▶ when they agree, the verdict is robust; when they disagree, look at the residual plot and at the sample
- In Python: `coint(y, x)` from `statsmodels`; `phillips_ouliaris(y, x, test_type="Zt")` from `arch`

Worked example: US consumption and income

- Data: c_t , $y_t = 100 \times \log$ of real consumption and real GDP, quarterly, 1959Q1–2009Q3 (203 quarters)
 - ▶ pretest: ADF with trend, $p = 0.31$ for c_t and 0.39 for y_t : both $I(1)$
- Step 1: $\hat{c}_t = -107.6 + 1.075 y_t$, $R^2 = 0.9992$, $DW = 0.24$
 - ▶ DW: the Durbin–Watson statistic of the residuals, about 2 without autocorrelation and close to 0 for very persistent residuals
 - ▶ the slope is close to 1: a stable consumption share; but R^2 and DW alone prove nothing
- Step 2: $\tau = -3.54$ with 0 lags; 5% critical value for $n = 2$, $T = 203$: -3.37
 - ▶ $-3.54 < -3.37$: reject “no cointegration” at 5% ($p = 0.029$); Phillips–Ouliaris: $Z_t = -3.52$, $p = 0.031$
 - ▶ with the Dickey–Fuller value (-2.88) the decision would be the same here, but for the wrong reason

The step-1 residuals



- Left and centre: $\hat{u}_t$ of consumption on income and of the 10-year on the 3-month Treasury yield; right: their sample ACF and that of a simulated random walk

Interpreting the residuals

- ▣ Consumption: $\hat{\rho}(1) = 0.86$ but $\hat{\rho}(8) = 0.31$: deviations of 1–3% that shrink markedly within about two years
- ▣ Yields: $\hat{\rho}(1) = 0.97$, $\hat{\rho}(12) = 0.58$: very persistent deviations; a borderline test ($\tau = -3.35$, $p = 0.048$)
 - ▶ high persistence is the usual situation: the power of the test against ρ close to 1 is low ([Chapter 3](#))
- ▣ A random walk decays much more slowly: the ACF alone cannot settle the question; the test can

Engle–Granger and Phillips–Ouliaris on real data

Regression	Freq.	T	$\hat{b}$	$\tau(k)$	p	Z_t	p
US consumption on income	quarterly	203	1.07	−3.54 (0)	0.029	−3.52	0.031
US: 10-year on 3-month yield	monthly	801	0.86	−3.35 (20)	0.048	−4.04	0.006
TLV on BRD (since 2014)	daily	3 176	1.36	−4.33 (3)	0.002	−4.14	0.005
BRD on TLV (since 2014)	daily	3 176	0.71	−4.16 (3)	0.004	−4.02	0.007
TLV on BRD (since 2010)	daily	4 187	1.68	−1.82 (3)	0.622	−1.83	0.619
EUR/RON on EUR/HUF	monthly	255	0.71	−2.47 (0)	0.294	−2.43	0.309
Romania: consumption on GDP	quarterly	126	1.46	−2.32 (8)	0.363	−6.80	< 0.001

- τ : Engle–Granger statistic (k lags by AIC); p -values of [MacKinnon \(2010\)](#) for 2 variables; Z_t : Phillips–Ouliaris
- Banks: adjusted prices from data/market; Romania: household consumption and GDP, Eurostat, chain-linked volumes, seasonally adjusted, since 1995

Interpreting the table

- **Normalisation:** TLV on BRD and BRD on TLV give different τ (-4.33 and -4.16) and slopes that are not exact inverses
 - ▶ Engle–Granger depends on which variable is on the left; here the verdict is the same
- **The sample:** since 2010 the same banks are not cointegrated ($p = 0.622$); since 2014 they are ($p = 0.002$)
 - ▶ choosing the start date after looking at the data is a form of data snooping
- **The two tests disagree** for Romania: Engle–Granger $p = 0.363$ with 8 lags, Phillips–Ouliaris $p < 0.001$
 - ▶ the slope 1.46 is far from 1: the consumption share rose strongly; Seminar 7 (B2) discusses this case

Limits of the Engle–Granger method

- ▣ **At most one** cointegrating vector: with $n \geq 3$ variables there may be several (Sections 4–5)
- ▣ **Normalisation**: the results depend on the variable put on the left
- ▣ **Inference on $\hat{b}$** : super-consistent, but biased in small samples, and its OLS t -statistics are **not** valid
 - ▶ remedy: dynamic OLS (DOLS) [Stock and Watson \(1993\)](#), adding leads and lags of Δx_t , gives valid standard errors
- ▣ **Two-step errors**: mistakes in step 1 carry over to step 2; the power is low against slow adjustment
 - ▶ the Johansen method (Section 5) estimates everything at once, by maximum likelihood

Case study: Engle and Granger (1987)



Robert Engle (2022); Nobel Prize in
Economics 2003

Photo: Jon Demske (2022); CC BY-SA 4.0;
[Wikimedia Commons](#)

- **Question:** how to model $I(1)$ series that are tied in the long run, without spurious regressions and without differencing away the long run?
 - ▶ [Engle and Granger \(1987\)](#), written at the University of California, San Diego, one of the most cited papers in econometrics
- **Contribution**
 - ▶ the definition of cointegration and the representation theorem (Section 4): cointegration $\Leftrightarrow$ error correction
 - ▶ the two-step estimator and residual-based tests with simulated critical values
 - ▶ applications to US series such as consumption and income, prices and wages, short and long interest rates

Recap: The Engle–Granger method

- Step 1: OLS in levels (super-consistent $\hat{b}$); step 2: ADF on the residuals, H_0 : no cointegration
- Critical values of Engle–Granger (MacKinnon), more negative than Dickey–Fuller and dependent on n
- Check with Phillips–Ouliaris, with the reverse normalisation and with another sample

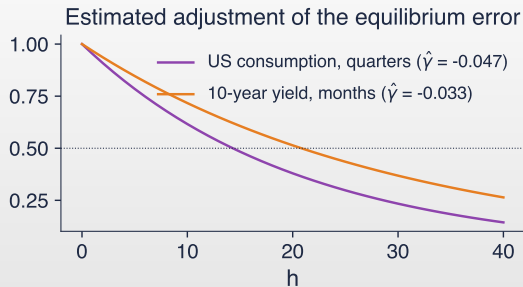
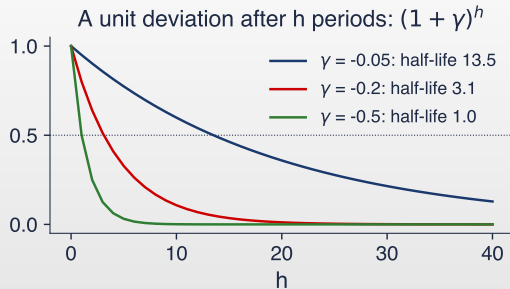
The error correction model

- With $u_{t-1} = y_{t-1} - a - bx_{t-1}$ (last period's equilibrium error), the **error correction model (ECM)** is
 - ▶ $\Delta y_t = c + \gamma u_{t-1} + \delta_0 \Delta x_t + \sum_{j \geq 1} (\phi_j \Delta y_{t-j} + \delta_j \Delta x_{t-j}) + \varepsilon_t$
- Every term is stationary: OLS and the usual t -tests are valid (with $\hat{u}_{t-1}$ from step 1)
 - ▶ b : the **long-run** effect of x on y ; δ_0 : the **short-run** (same-period) effect
 - ▶ γ : the **speed of adjustment**; error correction requires $-2 < \gamma < 0$
 - ▶ c : constant; ϕ_j, δ_j : short-run coefficients of the lagged differences; ε_t : white noise
- Reading: if y_{t-1} was above equilibrium ($u_{t-1} > 0$) and $\gamma < 0$, then y_t tends to fall: a share $|\gamma|$ of the gap is closed each period

Speed of adjustment and half-life

- Without new shocks, the equilibrium error evolves as $u_t = (1 + \gamma) u_{t-1}$, so $u_h = (1 + \gamma)^h u_0$
 - ▶ (in a bivariate system where only y adjusts)
- **Half-life:** the number of periods after which half of a deviation is gone
 - ▶ $(1 + \gamma)^h = 0.5 \Rightarrow h_{1/2} = \ln 0.5 / \ln(1 + \gamma)$
- **Worked example:** $\gamma = -0.2$, $h_{1/2} = \ln 0.5 / \ln 0.8 = 3.1$ periods
 - ▶ in months: about three months for half of a deviation; in quarters: three quarters
- Compare half-lives only in the same time unit: a monthly and a quarterly γ are not comparable

Half-lives: theory and estimates



- Left: $(1 + \gamma)^h$ for three speeds; right: the decay implied by the estimated ECMs of US consumption (quarters) and of the 10-year yield (months)

Interpreting the estimated ECMs

- ▣ Consumption: $\hat{\gamma} = -0.047$ ($t = -1.75$), half-life 14.3 quarters; $\hat{\delta}_0 = 0.54$
 - ▶ half of a rise in income reaches consumption in the same quarter; the rest comes slowly, and the adjustment term is only marginally significant
- ▣ 10-year yield: $\hat{\gamma} = -0.033$ ($t = -4.94$), half-life 20.8 months; $\hat{\delta}_0 = 0.38$
 - ▶ the 3-month rate also reacts to the same gap: $\hat{\gamma} = 0.024$ ($t = 2.08$), with the opposite sign: both rates move towards each other
- ▣ When both variables adjust, a single-equation ECM tells only half of the story: hence the VECM (Section 4)

Worked example: one quarter of consumption

- ▣ Estimated ECM (main terms): $\Delta c_t = 0.418 + 0.54 \Delta y_t - 0.047 \hat{u}_{t-1} + \dots$, with $\hat{u}_{t-1} = c_{t-1} - 1.075 y_{t-1} - \hat{a}$
- ▣ Scenario: consumption is 2% above its equilibrium ($\hat{u}_{t-1} = 2$) and income grows by 1% this quarter
 - ▶ short-run effect: $0.537 \times 1 = 0.537$; error correction: $-0.047 \times 2 = -0.095$
 - ▶ predicted growth of consumption: about 0.86% (the lagged differences are set to 0)
- ▣ Without the error correction term, the model would ignore that consumption is already too high

Case study: the consumption function of DHSY (1978)

- **Question:** why did the consumption functions of the 1970s, estimated in levels or in differences, forecast UK consumers' expenditure so badly?
 - ▶ [Davidson et al. \(1978\)](#) (Davidson, Hendry, Srba and Yeo, "DHSY"): quarterly UK data on expenditure and income
- **Method:** a model in differences plus the lagged log ratio of expenditure to income, the error correction term
 - ▶ a general-to-specific search, starting from a large dynamic model
- **Legacy:** the ECM form came first; [Granger \(1981\)](#) and [Engle and Granger \(1987\)](#) then showed why it works: the two series are cointegrated
 - ▶ a test of cointegration can also use the t -statistic of γ in the ECM, with its own critical values [Banerjee, Dolado and Mestre \(1998\)](#)

Recap: Error correction models

- ECM: Δy_t on Δx_t , lagged differences and $\hat{u}_{t-1}$; all terms stationary
- $\gamma < 0$: speed of adjustment; half-life $\ln 0.5 / \ln(1 + \gamma)$, in the units of the data
- Short run (δ_0) and long run (b) in one equation; if several variables adjust, we need a system

From a VAR to a VECM

- **Chapter 6:** a VAR(p) in levels, $\mathbf{y}_t = \mathbf{c} + A_1 \mathbf{y}_{t-1} + \cdots + A_p \mathbf{y}_{t-p} + \mathbf{u}_t$
 - ▶ subtract $\mathbf{y}_{t-1}$ and regroup the lags: the same model in **vector error correction** form (VECM)
- $\Delta \mathbf{y}_t = \mathbf{c} + \Pi \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t$, $\Pi = A_1 + \cdots + A_p - I$, $\Gamma_i = -(A_{i+1} + \cdots + A_p)$
 - ▶ Π : the long-run matrix (it multiplies the levels); Γ_i : the short-run matrices; I : the identity matrix
- **Derivation** for $p = 2$: $\mathbf{y}_t - \mathbf{y}_{t-1} = (A_1 - I)\mathbf{y}_{t-1} + A_2 \mathbf{y}_{t-2} + \mathbf{u}_t$; add and subtract $A_2 \mathbf{y}_{t-1}$:
 - ▶ $\Delta \mathbf{y}_t = (A_1 + A_2 - I)\mathbf{y}_{t-1} - A_2 \Delta \mathbf{y}_{t-1} + \mathbf{u}_t$: $\Pi = A_1 + A_2 - I$, $\Gamma_1 = -A_2$
- The VECM is a rewriting, not a new model: all the information about the long run sits in the matrix Π

The rank of Π

- ▣ $\Delta \mathbf{y}_t$ is stationary; so $\Pi \mathbf{y}_{t-1}$ must be stationary too. Three cases, by $r = \text{rank}(\Pi)$:
- ▣ $r = 0$: $\Pi = 0$; no cointegration; the right model is a **VAR in differences**
- ▣ $r = n$: Π has full rank; $\mathbf{y}_t$ is already stationary; a **VAR in levels** ([Chapter 6](#))
- ▣ $0 < r < n$: **cointegration** with r vectors; $\Pi = \alpha\beta^\top$ with α, β of size $n \times r$
 - ▶ $\beta^\top \mathbf{y}_{t-1}$: the r equilibrium errors; α : the **adjustment coefficients** (loadings), how strongly each variable reacts to each error
 - ▶ only the product is unique: $\alpha\beta^\top = (\alpha Q)(\beta Q^{-\top})^\top$ for any invertible Q ; we normalise β
- ▣ Testing for cointegration = testing the rank of Π (Section 5)

The Granger representation theorem

- ▣ **Theorem Engle and Granger (1987)**: if $\mathbf{y}_t$ is $I(1)$ and cointegrated with rank r , then it has a VECM representation with $\Pi = \alpha\beta^\top$, and conversely
 - ▶ cointegration $\Leftrightarrow$ error correction: the equilibrium error must feed back into the changes
- ▣ **Consequences**
 - ▶ a VAR in differences omits $\alpha\beta^\top \mathbf{y}_{t-1}$: it is **misspecified** when the series are cointegrated
 - ▶ $\alpha \neq 0$: at least one variable adjusts, so there is Granger causality ([Chapter 6](#)) in at least one direction
 - ▶ the levels are driven by $n - r$ common stochastic trends; the effects of shocks on the levels do not die out
- ▣ A VAR in levels is not wrong either: it is consistent, but it ignores the restriction $\text{rank}(\Pi) = r$ and its tests are non-standard

2003: the Nobel Prize for cointegration



Clive Granger (1934–2009)

Photo: [Olaf Storbeck \(2008\)](#); CC BY-SA 2.0;
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The Sir Clive Granger Building, University of Nottingham

Photo: [mattbuck \(2012\)](#); CC BY-SA 3.0; [Wikimedia Commons](#)

- The 2003 prize: Granger for cointegration, shared with Robert Engle for ARCH ([Chapter 5](#))
- Granger studied and began his career at Nottingham
- Nobel lecture: [Granger \(2004\)](#): “time series analysis, cointegration, and applications”

Worked example: a bivariate VECM

- $\Delta y_{1t} = -0.10(y_{1,t-1} - y_{2,t-1}) + u_{1t}, \quad \Delta y_{2t} = 0.10(y_{1,t-1} - y_{2,t-1}) + u_{2t}$
 - ▶ $\beta = (1, -1)^\top, \alpha = (-0.10, 0.10)^\top; \Pi = \alpha\beta^\top = \begin{pmatrix} -0.10 & 0.10 \\ 0.10 & -0.10 \end{pmatrix}, \det \Pi = 0, \text{rank } 1$
- If y_1 is above y_2 : y_1 falls ($\alpha_1 < 0$) and y_2 rises ($\alpha_2 > 0$): both close the gap
- The gap $z_t = y_{1t} - y_{2t}$: $z_t = (1 + \beta^\top \alpha) z_{t-1} + (u_{1t} - u_{2t}) = 0.8 z_{t-1} + \dots$
 - ▶ a stationary AR(1); half-life $\ln 0.5 / \ln 0.8 = 3.1$ periods; the condition is $-2 < \beta^\top \alpha < 0$
- The average $(y_{1t} + y_{2t})/2$ changes by $(u_{1t} + u_{2t})/2$: a random walk, the common trend

Deterministic terms: five cases

Case	In the VECM	Implication for the data	statsmodels
1	no constant, no trend	no drift; relations through the origin	n
2	constant only inside $\beta^\top y_{t-1}$	no drift; equilibrium errors with a non-zero mean (yields, spreads)	ci
3	unrestricted constant	linear trends in the levels, none in the relations (GDP, prices)	co
4	constant, and a trend inside $\beta^\top y_{t-1}$	trends in the levels and in the relations	coli
5	unrestricted constant and trend	quadratic trends in the levels; rarely plausible	colo

- The critical values of the Johansen tests change with the case [Johansen \(1995\)](#); choose the case from the plot and from theory, before testing
- `coint_johansen` of `statsmodels` implements case 1 (`det_order=-1`) and case 3 (`det_order=0`), the case used in this chapter

Weak exogeneity

- If a row of α is zero, $\alpha_i = 0$, the variable y_i does **not** react to the equilibrium errors
 - ▶ y_i is **weakly exogenous** for β : the other variables do all the adjusting
 - ▶ y_i pushes the common trend; e.g. Euribor for Romanian interest rates (Seminar 7, B4)
- Test: a t -test on α_i (one cointegrating vector) or a likelihood-ratio (LR) test of $\alpha_i = 0$ [Johansen \(1991\)](#)
 - ▶ the statistic is asymptotically χ^2 with r degrees of freedom, once r is fixed
- Use: if x_t is weakly exogenous, a single-equation ECM for y_t (Section 3) loses no information about β

Recap: The VECM

- ▣ $\Delta \mathbf{y}_t = \alpha \beta^\top \mathbf{y}_{t-1} + \sum \Gamma_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t$: a VAR rewritten; $\text{rank}(\Pi) = r$
- ▣ Granger representation: cointegration $\Leftrightarrow$ error correction; a VAR in differences omits the long run
- ▣ β : equilibria; α : who adjusts; deterministic terms: five cases; $\alpha_i = 0$: weak exogeneity

The idea: reduced-rank regression



University of Copenhagen, where Søren Johansen and Katarina Juselius developed the method

Photo: Per Meistrup (2011); CC BY-SA 4.0;
Wikimedia Commons

- Johansen (1988), Johansen (1991): maximum likelihood for the VECM with Gaussian errors
 - ▶ regress $\Delta \mathbf{y}_t$ and $\mathbf{y}_{t-1}$ on the lagged differences; keep the residuals R_{0t} and R_{1t}
 - ▶ find the combinations of R_{1t} most correlated with R_{0t} : the **canonical correlations**
- Their squares are the eigenvalues $1 > \hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_n \geq 0$
 - ▶ a large $\hat{\lambda}_i$: a stationary combination that predicts the changes;
 - $\hat{\lambda}_i \approx 0$: a combination that is still $I(1)$
 - ▶ the eigenvectors give $\hat{\beta}$; then $\hat{\alpha}$ follows by OLS

Trace and maximum-eigenvalue statistics

- **Trace test:** H_0 : rank $\leq r$, against rank $= n$; $\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i)$
 - ▶ it sums the information in the $n - r$ smallest eigenvalues: if they are all close to 0, the statistic is small and H_0 is kept
- **Maximum-eigenvalue test:** H_0 : rank $= r$, against rank $= r + 1$; $\lambda_{\text{max}}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
 - ▶ it uses only the next eigenvalue, $\hat{\lambda}_{r+1}$; T : the number of observations
- **Sequential procedure:** test $r = 0$, then $r \leq 1, \dots$; the rank is the first r whose H_0 is **not** rejected
 - ▶ the statistics are likelihood ratios, but their distributions are non-standard (functions of Brownian motions)
 - ▶ critical values depend on $n - r$ and on the deterministic case: [Osterwald-Lenum \(1992\)](#), [MacKinnon, Haug and Michelis \(1999\)](#)
- The two tests usually agree; if not, the trace test is preferred in most applied work

Worked example: from eigenvalues to statistics

- ▣ US yields at 1, 5 and 10 years, monthly, $T = 801$: $\hat{\lambda} = (0.0405, 0.0229, 0.0039)$
 - ▶ $-\ln(1 - \hat{\lambda}_i)$: 0.0414; 0.0231; 0.0039
- ▣ $r = 0$: trace = $801 \times (0.0414 + 0.0231 + 0.0039) = 54.8 > 29.80$: reject
 - ▶ max-eigenvalue: $801 \times 0.0414 = 33.2 > 21.13$: reject
- ▣ $r \leq 1$: trace = $21.7 > 15.49$: reject; $r \leq 2$: trace = $3.1 < 3.84$: do not reject
 - ▶ rank 2: two cointegrating vectors, one common trend
- ▣ Small eigenvalues still matter: they are multiplied by T

Johansen tests on three systems

System	T	k	$\lambda_{\text{trace}}(0)$	$\lambda_{\text{trace}}(1)$	$\lambda_{\text{trace}}(2)$	$\lambda_{\text{max}}(0)$	$\lambda_{\text{max}}(1)$	rank
US yields 1y, 5y, 10y	801	2	54.6	21.6	3.1	33.0	18.5	2
US output, consumption, investment	203	1	28.9	11.4	2.6	17.4	8.9	0
EUR/RON, EUR/HUF, EUR/PLN	255	1	24.2	7.9	2.7	16.3	5.2	0
5% critical value			29.80	15.49	3.84	21.13	14.26	

- Case 3 (unrestricted constant); k : lagged differences chosen by BIC on the VAR in levels; ranks by the trace test (the maximum-eigenvalue test gives the same ranks here)
- Yields: FRED, 1960–2026; output, consumption and investment: 100 ln, US 1959–2009; currencies: BNR, month-end, 2005–2026

Interpreting the three systems

- Yields: rank 2, so one common trend: the **level** of rates moves all maturities; the two spreads are stationary
 - ▶ the expectations hypothesis of the term structure [Campbell and Shiller \(1987\)](#) in its weak form
- Output, consumption and investment: trace 28.9 against 29.80: rank 0 at 5%, a near miss
 - ▶ balanced growth predicts rank 2 (stable shares of consumption and investment); with Engle–Granger, consumption and income were cointegrated ($p = 0.029$): low power, not proof of absence
- Currencies: trace 24.2: rank 0; three separate stochastic trends (Section 8)

The deterministic case changes the answer

Deterministic terms	$\lambda_{\text{trace}}(0)$	5%	$\lambda_{\text{trace}}(1)$	5%	$\lambda_{\text{trace}}(2)$	5%
none (det_order=-1)	45.9	24.28	13.3	12.32	0.4	4.13
constant (det_order=0)	54.6	29.80	21.6	15.49	3.1	3.84
linear trend (det_order=1)	60.0	35.01	23.7	18.40	4.8	3.84

- US yields at 1, 5 and 10 years: ranks 2, 2 and 3
- With a linear trend, the test says that every yield is stationary around a trend: not plausible for interest rates over 66 years
 - choose the deterministic terms from theory and from the plot, not from the result you like
- Other practical issues: the lag length k ; size distortions in small samples (correction factor $(T - nk)/T$, [Reinsel and Ahn \(1992\)](#)); structural breaks

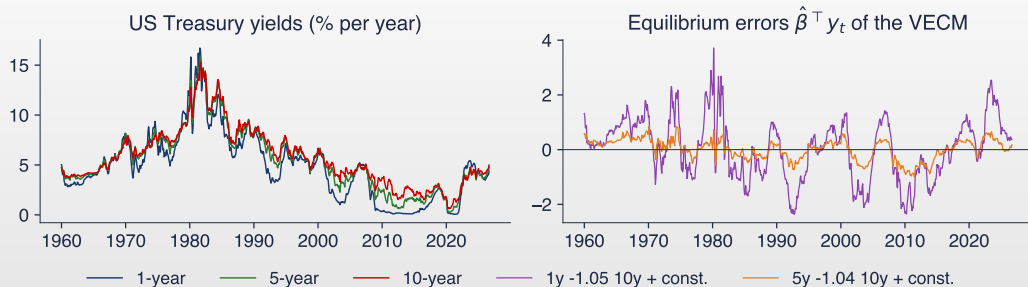
Case study: Johansen and Juselius (1990)

- **Question:** is there a stable demand for money in Denmark and Finland?
 - ▶ [Johansen and Juselius \(1990\)](#): real money, real income, interest rates (and inflation); quarterly data
- **Method:** the VECM by maximum likelihood; the rank by the trace and maximum-eigenvalue tests; then **tests of restrictions** on β and α
 - ▶ e.g. a unit income elasticity, $\beta = (1, -1, \cdot)$; weak exogeneity of income and interest rates
- **Legacy:** the template of the “cointegrated VAR” methodology [Juselius \(2006\)](#): test the rank, then test economic hypotheses as restrictions
 - ▶ one of the most cited papers in applied econometrics

Recap: The Johansen tests

- Eigenvalues of a reduced-rank regression; trace and maximum-eigenvalue statistics; sequential testing from $r = 0$
- Critical values depend on $n - r$ and on the deterministic case
- Yields: rank 2; macro aggregates: borderline; currencies: rank 0

The term-structure VECM



- US Treasury yields at 1, 5 and 10 years (FRED GS1, GS5, GS10), January 1960–September 2026; VECM with rank 2, 2 lagged differences, constant restricted to the relations (case 2)

Interpreting the cointegrating vectors

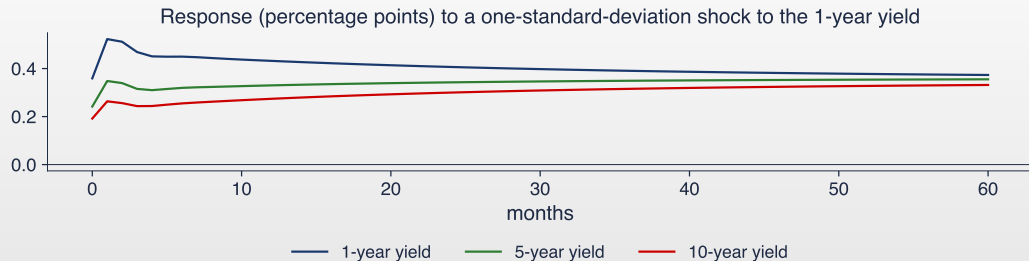
- ▣ Normalisation on the 1-year and 5-year yields: $\hat{\beta}_1^\top \mathbf{y}_t = i_t^{(1)} - 1.045 i_t^{(10)} + 1.22$, $\hat{\beta}_2^\top \mathbf{y}_t = i_t^{(5)} - 1.044 i_t^{(10)} + 0.59$
 - ▶ both slopes close to 1: the equilibria are almost the spreads $i^{(1)} - i^{(10)}$ and $i^{(5)} - i^{(10)}$, with mean term premia
- ▣ The equilibrium errors are stationary: ADF $p < 0.001$ and 0.008 ; they are large in the 1980s and around zero-rate periods
- ▣ A formal test of $\beta = (1, -1)$ for each spread is a likelihood-ratio test [Johansen \(1991\)](#); with super-consistent $\hat{\beta}$, values this close to 1 are informative

The adjustment coefficients

Equation	$\hat{\alpha}_{.1}$	t	$\hat{\alpha}_{.2}$	t
$\Delta i_t^{(1)}$ (1 year)	-0.021	-0.82	0.024	0.31
$\Delta i_t^{(5)}$ (5 years)	0.045	2.21	-0.122	-2.00
$\Delta i_t^{(10)}$ (10 years)	0.040	2.22	-0.077	-1.43

- The 1-year yield does not react to either error ($|t| < 1$): it behaves as **weakly exogenous**
 - the short end follows monetary policy; the 5- and 10-year yields do the adjusting ($|t| \approx 2$)
- The equilibrium errors $\mathbf{z}_t = \hat{\beta}^\top \mathbf{y}_t$ follow $\mathbf{z}_t = (I + \hat{\beta}^\top \hat{\alpha}) \mathbf{z}_{t-1} + \dots$; eigenvalues 0.969 and 0.926
 - half-lives of about 22.1 and 9.0 months: slow error correction, typical of interest rates

Impulse responses of the VECM



□ Orthogonalised responses (Cholesky, ordering 1y, 5y, 10y; [Chapter 6](#)) to a one-standard-deviation shock to the 1-year yield

 TSA_ch7_johansen_vecm

Interpreting the impulse responses

- On impact: +0.36, +0.24 and +0.19 percentage points for 1, 5 and 10 years
 - ▶ after 60 months: +0.37, +0.36 and +0.33: the effects do **not** die out and become almost equal
- A permanent shift of the common trend (the level), with stationary spreads: the signature of $n - r = 1$ common trend
- In a stationary VAR ([Chapter 6](#)) every response returns to zero; in a VECM the long-run responses are generally different from zero

Recap: β and α

- ▣ $\hat{\beta}$: normalise, read as equilibria (spreads, ratios), test economic restrictions
- ▣ $\hat{\alpha}$: who adjusts and how fast; a zero row means weak exogeneity
- ▣ Shocks have permanent effects on cointegrated levels, but not on the equilibrium errors

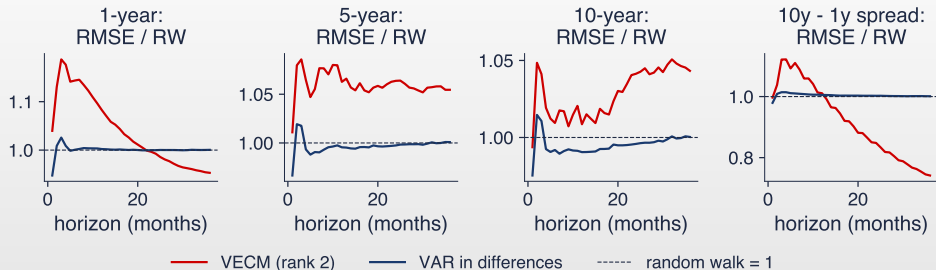
The theoretical arguments

- VECM forecasts keep the equilibria: as the horizon grows, $\hat{\beta}^\top \hat{\mathbf{y}}_{T+h}$ returns to its mean
 - ▶ a VAR in differences forgets the levels: its forecasts of the spreads stay where they are
- Engle and Yoo (1987): imposing cointegration improves long-horizon forecasts in simulated systems
- Christoffersen and Diebold (1998): the gains are concentrated in the **cointegrating combinations**
 - ▶ for the individual levels, the forecast error is dominated by the common trend, which no model can predict
 - ▶ estimation error in α , β and in the constants can make a VECM worse in practice
- Test it out of sample, with re-estimation at each forecast origin

The forecasting experiment

- ▣ Data: US yields at 1, 5 and 10 years, monthly, since 1960
- ▣ 135 forecast origins, every three months from January 1990 to July 2023; models re-estimated on the data up to each origin
 - ▶ VECM with rank 2 and a restricted constant; VAR in differences without constant, with the same lags; random walk (no change)
- ▣ Horizons of 1 to 36 months; RMSE (root mean squared error) of each model divided by that of the random walk
 - ▶ ratio below 1: better than the random walk; also for the 10-year minus 1-year spread

VECM against a VAR in differences, out of sample



□ RMSE relative to the random walk by horizon; first three panels: the yields; last panel: the 10-year minus 1-year spread

Interpreting the forecast comparison

- ▣ Levels: neither model beats the random walk clearly; the VECM is worse at short horizons (1 year at 6 months: 1.14; 5 years: 1.06)
 - ▶ the VAR in differences stays close to 1 at every horizon (1 year at 12 months: 1.00)
- ▣ Spread: the VECM gains at long horizons: 0.85 at 24 months and 0.74 at 36 months; the VAR in differences: 1.00
 - ▶ exactly the pattern of [Christoffersen and Diebold \(1998\)](#): cointegration helps forecast the equilibrium errors, not the trend
- ▣ Lesson: use a VECM when the object of interest is the relation (a spread, a ratio, a pair); for the levels alone, a random walk is hard to beat

Recap: Forecasting

- ▣ The VECM pulls the forecasts of the equilibrium errors back to their means; the VAR in differences does not
- ▣ Out of sample: gains for the spread at long horizons, none for the levels
- ▣ Always compare with the random walk, with re-estimation and without look-ahead

Purchasing power parity

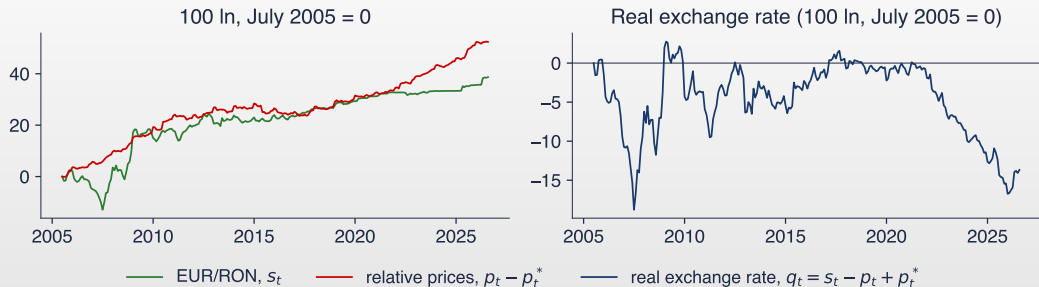


The National Bank of Romania, which publishes the reference rates

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons

- **Relative PPP** (purchasing power parity): the exchange rate offsets inflation differences
 - ▶ $s_t = 100 \ln(\text{EUR}/\text{RON})$; p_t, p_t^* : 100 $\ln$ of the HICP (Harmonised Index of Consumer Prices) of Romania and of the euro area
 - ▶ PPP holds in the long run if the real exchange rate $q_t = s_t - p_t + p_t^*$ is stationary: cointegration with $\beta = (1, -1, 1)$
- Reasons why it may fail
 - ▶ **Balassa (1964)**: fast productivity growth in tradables raises prices of non-tradables in a catching-up economy: a trend real appreciation
 - ▶ even where PPP holds, deviations decay slowly; the survey **Taylor and Taylor (2004)** reports half-lives of several years

PPP for the leu

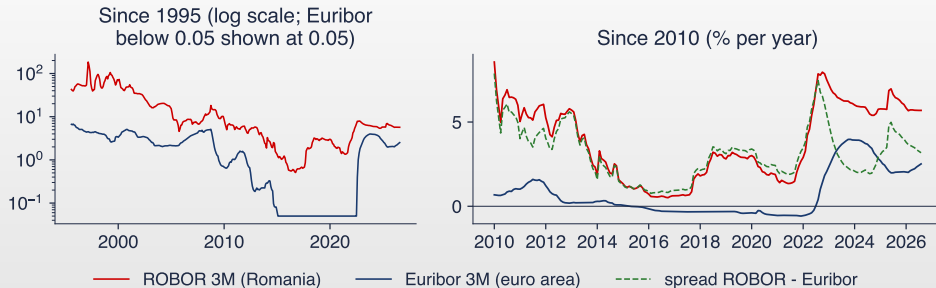


- Monthly, July 2005–August 2026: EUR/RON (monthly mean of the BNR reference rate), HICP of Romania and of the euro area (Eurostat, 2015 = 100)

Interpreting PPP for the leu

- ▣ Since 2005 the leu lost 38.7 log points against the euro, while Romanian prices rose 52.4 log points more than euro-area prices
 - ▶ the real exchange rate changed by -13.7 log points: a real appreciation of the leu, concentrated after 2021
- ▣ ADF on q_t : $p = 0.24$; Engle–Granger of s_t on $p_t - p_t^*$: slope 0.89, $p = 0.34$; Johansen on (s_t, p_t, p_t^*) : rank 0
- ▣ No evidence of PPP over 21 years, consistent with Balassa–Samuelson convergence; 21 years may also be too short for half-lives of several years

ROBOR and Euribor

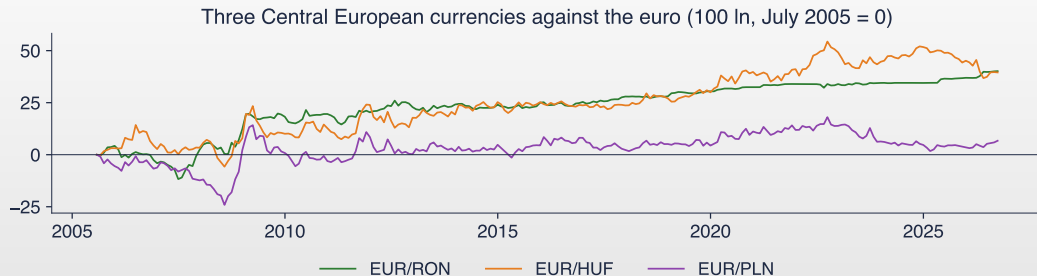


- ☐ Three-month money market rates (Eurostat), August 1995–August 2026: ROBOR 3M for Romania and Euribor 3M for the euro area

Interpreting ROBOR and Euribor

- ▣ Full sample: Engle–Granger $p = 0.52$, Phillips–Ouliaris $p = 0.010$: the disinflation of the 1990s dominates and the tests disagree
- ▣ Since 2010: $p = 0.047$ (Engle–Granger) and 0.029 (Phillips–Ouliaris), slope 1.19 , mean spread 3.11 percentage points
 - ▶ ECM for ROBOR: $\hat{\gamma} = -0.020$ ($t = -1.18$), half-life about 34 months: weak and slow adjustment
- ▣ Romanian monetary policy follows its own inflation target; the link to Euribor is loose (Seminar 7, B4, adds the Romanian 10-year yield)

Three Central European currencies



EUR/RON, EUR/HUF and EUR/PLN, cross rates from the BNR reference rates, month-end, 100 ln

Interpreting the three currencies

- ▣ Since July 2005: EUR/RON +40.1, EUR/HUF +39.5, EUR/PLN +6.7 log points; each rate is $I(1)$ (ADF $p = 0.55; 0.75; 0.08$)
- ▣ Johansen: trace 24.2 for $r = 0$ (critical value 29.80); also no cointegration since 2012 (trace 12.7)
 - ▶ a common anchor (the euro) and similar shocks do not create a common trend: each central bank and each inflation path adds its own
- ▣ The right model for these rates is a VAR in differences ([Chapter 6](#)); Seminar 7 (B3) repeats the analysis

Recap: Economic examples

- PPP for the leu is rejected over 2005–2026: real appreciation of a converging economy
- ROBOR and Euribor: a loose, slow link since 2010; the sample decides the verdict
- Theory proposes β ; the tests, the sample and the deterministic terms decide whether the data agree

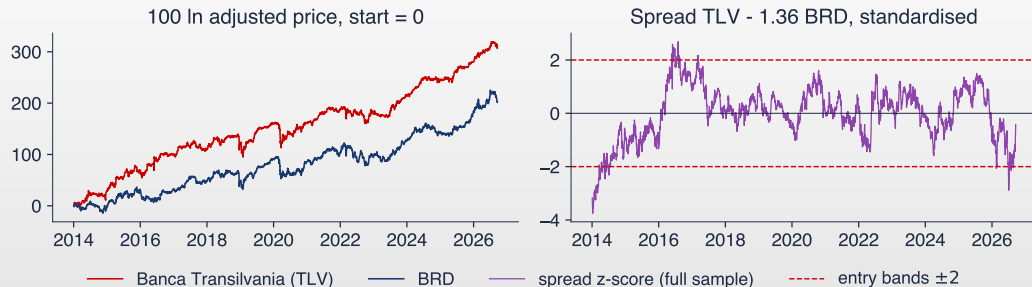
The idea of pairs trading

- Two shares exposed to the same shocks (two banks, two oil companies) may be cointegrated: their spread mean-reverts
 - ▶ when the spread is unusually wide, sell the expensive share and buy the cheap one; close when the spread is back
- A **relative-value** trade: the market direction cancels out; the profit comes from the error correction
 - ▶ Gatev, Goetzmann and Rouwenhorst (2006): the distance method on US shares, profitable over 1962–2002
 - ▶ Do and Faff (2010): the profits declined in later years
- A good test of a time-series idea: it must work **out of sample**, **after costs**, and on pairs chosen **without looking ahead**

A rule without look-ahead

- ▣ **Formation** (the previous 252 trading days): test every pair with Engle–Granger; keep the pairs with $p < 0.05$
 - ▶ $P_{A,t}$, $P_{B,t}$: the prices of shares A and B; r_A , r_B : their daily log returns
 - ▶ hedge ratio $\hat{b}$ by OLS of $\ln P_A$ on $\ln P_B$; mean $\bar{s}$ and standard deviation $\hat{\sigma}_s$ of the spread
$$s_t = \ln P_{A,t} - \hat{a} - \hat{b} \ln P_{B,t}$$
- ▣ **Trading** (the next 126 days), with the formation parameters fixed: $z_t = (s_t - \bar{s})/\hat{\sigma}_s$
 - ▶ $z_t > 2$: short A, long $\hat{b}$ of B; $z_t < -2$: the opposite; close at $z_t = 0$ or at the end of the window
 - ▶ daily return per unit of gross exposure: $\pm(r_A - \hat{b} r_B)/(1 + |\hat{b}|)$; equal capital per selected pair
- ▣ **Costs**: 0.20% (Bucharest) and 0.05% (US) of the gross exposure traded, at each opening and closing

Banca Transilvania and BRD

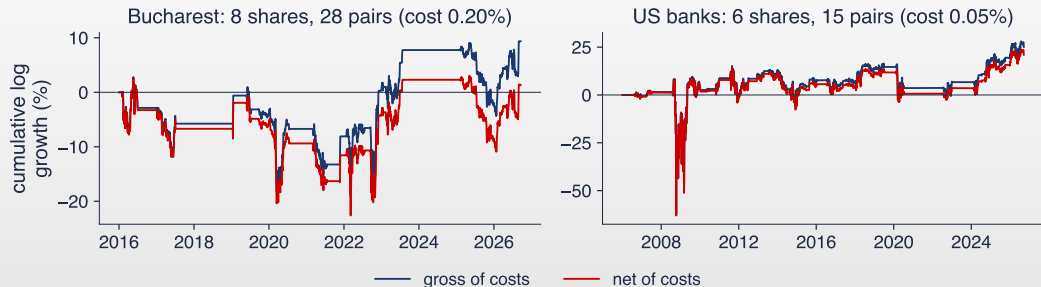


- Adjusted daily prices since 2014; right: the spread of the full-sample regression, standardised, with the entry bands

Interpreting the bank pair

- ▣ Full sample: $\hat{b} = 1.36$; the spread is an AR(1) with $\hat{\rho} = 0.990$: half-life about 69 trading days
 - ▶ only 5.0% of the days lie outside ± 2 : few trading opportunities, each lasting months
- ▣ Trading this picture with full-sample parameters gives 3.32% per year (Sharpe ratio 0.44), after costs
 - ▶ Sharpe ratio: the mean annual return divided by the annual standard deviation of the returns
 - ▶ an illusion: $\hat{b}$, $\bar{s}$ and $\hat{\sigma}_s$ use the future, and the pair was chosen because it looks cointegrated since 2014
- ▣ With the rolling rule, the pair passes the formation test in none of its 24 windows: one year of data is too short to detect such slow error correction

Pairs trading out of sample



- Left: 8 liquid BVB shares (TLV, BRD, SNP, TGN, TEL, EL, SNG, SNN), since 2015; right: 6 US banks (JPM, BAC, C, WFC, GS, MS), since 2005

Interpreting the backtest

- Bucharest: 22 windows, pairs found in 14, 73 trades, 62% of trades profitable, but the mean return per trade is -0.26%
 - ▶ gross 1.17% per year (Sharpe 0.15); net 0.41% (Sharpe 0.05); break-even cost 0.31%
- US banks: gross 2.77% (Sharpe 0.16), net 2.57% (Sharpe 0.14), 80 trades
 - ▶ with a deep loss in the autumn of 2008: cointegration can break exactly when it matters
- Many small wins and a few large losses: the typical profile of a mean-reversion strategy

Pitfalls of pairs trading

- ▣ **Multiple testing:** with 28 pairs and a 5% test, about 1.4 pairs pass by chance in each window
- ▣ **Look-ahead:** choosing pairs, start dates or thresholds after seeing the whole sample
- ▣ **Breaks:** mergers, recapitalisations, regulation and crises change β ; the spread may never come back
- ▣ **Market frictions:** short selling is limited on the BVB, borrowing shares costs a fee, bid-ask spreads are wide for small caps
- ▣ **Slow error correction:** half-lives of months mean capital tied up for months, and a long wait in a losing position

Recap: Pairs trading

- ▣ Pairs trading bets on error correction in the spread of two cointegrated shares
- ▣ In sample it looks attractive; out of sample, after costs, the profits on the BVB are close to zero
- ▣ Report the selection, the costs, the number of trades and the worst episode, not only the Sharpe ratio

Possible contribution of AI

- ▣ **Code:** a first draft of a script that runs unit-root tests, Engle–Granger, Phillips–Ouliaris and Johansen on many series and tabulates the verdicts
- ▣ **Explanation:** a second explanation of the Granger representation theorem, or of a VECM output table
- ▣ **Exploration:** candidate pairs on the BVB, or parity conditions for several Central European countries
- ▣ Example prompt
 - ▶ Write Python code that loads monthly US Treasury yields GS1, GS5 and GS10 from FRED, runs the Johansen trace test with a constant and BIC lags, fits a VECM with the chosen rank using statsmodels, and prints beta normalised on the first variables, alpha with t-statistics and the half-lives of the equilibrium errors.

Checks you must run

- ▣ The critical values: Engle–Granger (MacKinnon) for residuals, not Dickey–Fuller; Johansen for the right deterministic case
- ▣ The direction of the null: “no cointegration” in Engle–Granger; “rank $\leq r$ ” in the trace test
- ▣ That every series is $I(1)$ before testing, and that a high correlation of levels is not reported as cointegration
- ▣ The sign of the adjustment coefficients and the half-lives, in the right time unit
- ▣ Backtests: no look-ahead in pair selection, costs included, the number of pairs tested reported
- ▣ Every cited reference: it must exist; check the DOI

Key takeaways

- ▣ Cointegrated $I(1)$ series share common stochastic trends; a combination $\beta^\top \mathbf{y}_t$ is stationary
- ▣ Engle–Granger: OLS in levels, then ADF on the residuals with MacKinnon critical values; check with Phillips–Ouliaris
- ▣ ECM: changes respond to the lagged equilibrium error; γ gives the speed and the half-life
- ▣ VECM: $\Pi = \alpha\beta^\top$; Granger representation: cointegration $\Leftrightarrow$ error correction; Johansen tests choose r
- ▣ Cointegration helps forecast relations (spreads), not levels; applications must be judged out of sample and after costs

Key formulas

Quantity	Formula
Cointegration	$\mathbf{y}_t \sim I(1), \quad \beta^\top \mathbf{y}_t \sim I(0)$
Engle–Granger	$y_t = a + b x_t + u_t; \quad \Delta \hat{u}_t = \gamma \hat{u}_{t-1} + \sum_j \delta_j \Delta \hat{u}_{t-j} + e_t, \quad H_0: \gamma = 0$
ECM	$\Delta y_t = c + \gamma \hat{u}_{t-1} + \delta_0 \Delta x_t + \dots + \varepsilon_t, \quad h_{1/2} = \ln 0.5 / \ln(1 + \gamma)$
VECM	$\Delta \mathbf{y}_t = \alpha \beta^\top \mathbf{y}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t, \quad \Pi = \sum_i A_i - I, \quad \Gamma_i = -\sum_{j>i} A_j$
Johansen	$\lambda_{\text{trace}}(r) = -T \sum_{i>r} \ln(1 - \hat{\lambda}_i), \quad \lambda_{\text{max}}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
Equilibrium errors	$\mathbf{z}_t = \beta^\top \mathbf{y}_t; \quad \mathbf{z}_t = (I + \beta^\top \alpha) \mathbf{z}_{t-1} + \dots$

Self-assessment

- ▣ **Question:** Engle–Granger with two variables gives $\tau = -3.1$ for $T = 300$. Do you reject “no cointegration” at 5%?
 - ▶ **Answer:** no: $-3.1 > -3.36$; with the Dickey–Fuller value (-2.87) you would wrongly reject
- ▣ **Question:** an ECM gives $\hat{\gamma} = -0.1$ on monthly data. What is the half-life?
 - ▶ **Answer:** $\ln 0.5 / \ln 0.9 \approx 6.6$ months
- ▣ **Question:** four $I(1)$ variables have cointegration rank 3. How many common trends drive them?
 - ▶ **Answer:** $n - r = 1$
- ▣ Next: [Chapter 8](#), long memory: series between $I(0)$ and $I(1)$, fractional differencing and ARFIMA

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